Alex Tefft

Project 2

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In popular culture, the computer scientist is often portrayed alone in a dark room typing away at a computer console. Within the actual discipline, this couldn't be further from the truth. Computer Science is more than just the study of machines and computers; it is a community of practice where writing functions as the connections between humans and computers. Whether through the syntax of a Java file or the words in a peer-reviewed paper, writing in CS is a dynamic, multimodal act that establishes professional legitimacy and mediates the progress of science itself.

As I explored through interviews with an instructor and a peer, the transition into this field involves a shift in identity. One must move away from a "problem-solving" mindset learned in school (what my instructor interviewee called the "gold star" approach, for the "gold star" stickers teachers may give out) toward a communication mindset, where the value of a solution is determined by how effectively it can be

understood, critiqued, and built upon by others. By examining these perspectives alongside threshold concepts, this paper will illustrate how writing in Computer Science reflects shared values and supports the collective participation necessary for the field to evolve. To see how these values translate into practice, I will then focus on the Scrum development team, where writing serves as the primary tool for collaborative software production.

If writing is a “dynamic, multimodal technology for making meaning,” as the threshold concept suggests, then Computer Science is perhaps the ultimate example of this definition. However, what actually counts as “writing” depends on who you ask within the community. For the peer I interviewed, the definition is broad and functional: "code is definitely writing to me." Through this view, Python or C++ are simply languages with a unique audience: the computer itself. My instructor, however, observed a divergence within the field’s genres. He noted that while a software package or block of code provides the functional implementation of a solution, the academic paper is a separate layer that links to the code rather than being a translation of it. This is an important note that suggests writing in CS is not a single act, but a multimodal stack. At the bottom layer, code mediates the activity between programming and machine, while at the top, scholarly papers mediate the activity between researchers and the greater scientific community. To be a practitioner of Computer Science is to be a translator between all of these modes, ensuring both that the machine executes the logic and the human community understands its significance.

This layered understanding of writing in CS directly relates to two threshold concepts: writing is a dynamic multimodal technology for making meaning and writing

mediates activity. Writing is the code a machine reads, the documentation that the developers reference, the papers the researcher publishes, and even the pull request comments reviewers of code leave. Each of these modes mediates a different layer of activity, from the programmer and machine interaction at the base to the researcher and community interaction at the top. What makes Computer Science unique compared to other disciplines is that computer scientists must be fluent across all layers of this stack simultaneously; A solution that only works at the code level is meaningless if it cannot be communicated effectively at the scholarly level, and vice versa.

While the medium of writing may shift, the underlying values of the discipline seem to be focused on precision and transparency. In Computer Science, these are more than just stylistic preferences, and are instead fully functional. My peer interviewee emphasized that in code, precision is vital. If the code isn't logically sound, the "audience" (computer) will fail to respond in the way the programmer is looking for. However, this value extends to the social interactions of the field as well. My peer talked about their time working at an internship and noted that comments on his pull requests (essentially formal requests to merge new code into a main project) initially felt angry or personal. He soon realized that the bluntness was a reflection of the community's values. The goal is "functioning code," over the protection of the writer's ego. This aligns with Carolyn Miller's argument that genre is "centered not on the substance or the form of discourse but on the action it is used to accomplish" (Miller 151). In this case, the code review is not just a checklist making sure the code fulfills each requirement, but is a full genre with its own rhetorical purposes. First, it enforces community norms by holding code to shared standards of readability, efficiency, and correctness. It also

shapes collaboration by structuring the ways developers suggest improvements to each other's work. The reviewers' comments are formal documented contributions that become part of a project's record. Finally, it signals expertise. A well-written review demonstrates the reviewers' understanding of the codebase and establishes their credibility within the team. The genre's social action is the maintenance and expansion of the program's functionality, and it achieves this by encouraging a transparent, egoless communication style where the quality of the code takes priority over the feelings of the writer.

The transition from a student to a professional in Computer Science is marked by a shift in how one defines 'success' in writing. My instructor described this as moving away from the 'gold star' approach of k-12 and early college, where the goal is simply to solve the problem to get credit. This mirrors the findings of Johnson and Krase, who observe that graduating seniors often point to "the importance of junior- and senior-level tasks to their development as writers" (2). They suggest that for a student to truly "come to learn" as a writer, they must move beyond a "paper-for-grade economy" (10). As my instructor noted, "you can solve a problem, but it won't matter if you can't communicate that you have solved it." This realization marks the point where a student stops "simplifying a writing task" to make it accessible for a grade and begins to realize their own ideas as part of a "coherent field of diverse voices" (Johnson and Krase 2, 14). This illustrates the threshold concept that writing enacts disciplinarity. The act of writing the 'story' of the code is the act of becoming a scientist. By moving away from the 'gold star' mindset, the student begins to use writing to expand professional disciplinary knowledge rather than just satisfying a rubric. Writing in a discipline is a way of "coming

to learn." The writing does not just report the result, but also constructs the significance of the result. Writing moves from a passive "I found this" to an active, often persuasive argument for why the discovery matters to the wider scientific ecosystem.

If writing in Computer Science includes writing the "story" of a solution, then that story is frequently designed to provoke a specific response. This is related to the threshold concept that writing mediates activity. My instructor interviewee noted that he enjoys a more "aggressive" writing style, especially when relating to a new software package. He describes this genre as often inherently persuasive. Writers are essentially saying "My package is better; here is why you should use it." This is also an illustration of Miller's "Genre as Social Action." In this context, a software writeup or a research paper is a social tool used to convince a community to adopt a new algorithm, shift their methodology, or "act in some manner" to advance the field. Writing mediates the activity between the creator's discovery and the community's adoption. Without a rhetorical 'push', even the most useful code would remain stationary and never cross from being a private project to a public contribution.

Finally, writing serves as a representation of identity, where the quality of the writing or code directly constructs the author's professional persona. In computer science, your writing is your proxy. It is the primary way you establish what my instructor calls "legitimacy and competence." He pointed out that the consumer of your writing uses the clarity or precision of your writing to judge the quality of your mind. "Your writing defines you," my instructor noted. This creates a high-pressure environment for the writer. My peer agreed with this sentiment, and admitted he felt a sense of anxiety when writing for a mathematical modeling class. He was worried that if he was

perceived as a “bad communicator,” then others would be led to believe they didn’t understand the underlying math, even if the calculations were perfect. This reveals something about the discipline: to be seen as a legitimate computer scientist, one must be able to write themselves into that role, proving that identity as a computer scientist is linked to the CS documents we produce.

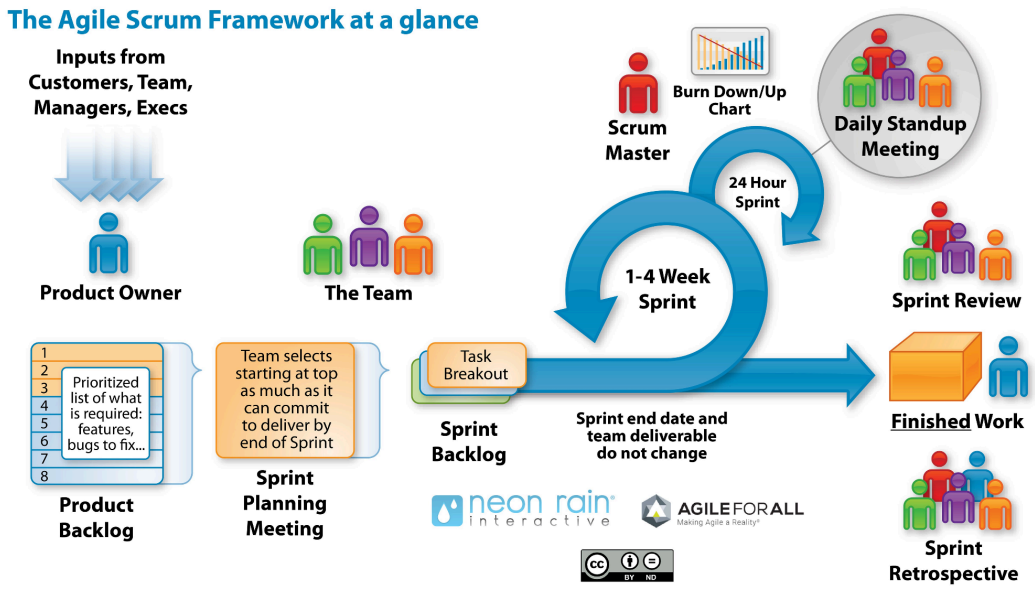
It has become clear to me that Computer Science is not the silent, purely logical practice it is often perceived to be. Instead it is a community where writing in many forms, whether via code, documentation, or scientific papers, serves as the primary vehicle for participation and progress. Writing is the tool that transforms a solution into science.

This project has started to reshape my understanding of my own identity within the field. Up until now, I have also had the “gold star” approach where most of my code is meant to comply with a strict rubric and metrics for a grade. By better understanding the values and ideologies that govern computer science writing now, I am more prepared to enter into the discipline through writing. Ultimately, I have learned that to be a successful computer scientist, one must master the ‘full stack’ of computer science writing as much as being a master of code syntax.

Understanding the general values of Computer Science is only the first step to becoming a Computer Scientist. To move from the ‘gold star’ reward system of the classroom to a more professional real-life working environment, it is important to be able to navigate specific sub-disciplines that each have their own rigid rules of engagement. By focusing on the specific discourse community of a Scrum team, we can see how writing functions not just as a reflection of the author’s identity, but is also a vital

mediating tool for creating software.

"Scrum" is a project management framework that divides work into short iterations called "sprints," usually lasting a few weeks. Unlike traditional methods where all of the requirements for the code are set from the beginning, Scrum acknowledges that software requirements will emerge as development continues. This necessitates constant written communication between the people involved. I am a newcomer to this community, although I have experienced Scrum through group projects in my Software Engineering classes, so I am still learning exactly how to navigate all of the specific rituals of Scrum teams. By analyzing this community through Swales' and Gee's frameworks and the lens of activity theory, I will argue that writing in a Scrum team is the primary mediating tool that structures collaboration, establishes authority, and enforces the shared social norms that allow a team to produce working software. Writing is not just a recording of the work a Scrum team has completed. It is the mechanism through which the work happens.



To understand how a Scrum team operates as a community, we can view it through the lens that Swales presents in “The Concept of Discourse Community” (1990) and his concept of the discourse community. Swales notes that such communities have a broadly agreed upon set of goals and methods of communication between members. For a Scrum team, the primary goal is the delivery of a working software that follows all requirements from the product owner.

The mechanisms of communication employed by a Scrum team are often digital and follow specific formal formats. This community uses specialized genres that include the “user story,” the “sprint backlog,” “acceptance criteria” and more. Each genre serves a very specific purpose. A user story, for example, is typically always written in the format "As a [type of user], I want [some goal] so that [some reason]." The purpose of these messages is not to fully detail the implementation of a feature, but to facilitate a conversation between a business and a developer. The community employs a variety of highly specialized terms like “story points” “velocity” “standup meeting” and more. Without an understanding of these genres and lexis, a developer would simply not be able to participate in the sprint.

These genres do more than just gate participation. They also actively shape how the team makes decisions and structures its work. The format of the user story, for example, constrains what counts as a valid feature request by forcing the author to articulate a user type, goal, and reason. This structure prevents value or otherwise technically infeasible requests from being submitted, and focuses the conversation on user value rather than technical implementation. The sprint backlog, meanwhile, functions as a contract. By committing to a specific set of stories for the sprint, the team

is collectively staking their credibility on delivering the features. Acceptance criteria serves a similar purpose. It defines the conditions which must be fulfilled for a user story to be considered ‘done,’ which removes ambiguity and gives both developers and stakeholders a shared standard to evaluate the work. Together, these genres structure the process of collaboration and decision-making within the Scrum team.

Finally, Swales notes that a discourse community also must have “a threshold level of members with a suitable degree of relevant content and discoursal expertise.” In a Scrum team, one must both possess the technical ability to code as well as the ability to participate in and navigate Scrum’s specific communication styles. Even if a developer is a coding genius, they will remain an outsider as long as they are not following the formats of communication laid out by Scrum. There are consequences to failing to adopt these conventions that go beyond just social ones. A developer who writes vague or incomplete user stories causes confusion during sprint planning, leading to timelines which may be estimated incorrectly or incomplete features. During a standup meeting, if a developer gives a longwinded technical monologue instead of the expected concise update (‘what I did, what I'm doing, what is blocking me’), they will waste the team's time and disrupt the expected rhythm of the shared ritual. Most importantly, a team member who fails to document blockers or update the sprint backlog creates information gaps that can cascade into a failed sprint. In each case, the inability to participate in Scrum's discourse conventions does not just make someone look out of place, but actively undermines the team’s productivity.

James Paul Gee’s framework allows us to examine the underlying dynamics of language, identity, and power that affect the discourse. Gee argues that entering a

discourse involves more than just skill (both technical and discoursal). It requires adopting a specific identity or a “way of being in the world” that signals membership to a community (Gee 6). In a Scrum team, this means that individual identity is secondary to the “development team” identity that developers take on, where the group acts as a single unit. In a Scrum team, the language difference between ‘we’ and ‘I’ is a very important membership signal. A developer who says, “we shipped the feature” is performing the identity of a team member, while one who says, “I finished all my work” inadvertently positions themself as an outsider. The Scrum framework reinforces this through its rituals, including standup meetings, sprint retrospectives, and demonstrations, which are collective activities where individual contributions become a team’s output. It is through these repeated linguistic practices that the team-based identity is enforced. Part of being a Scrum team member is speaking ‘the language of scrum,’ which is a language focused on ‘we’ instead of ‘I.’ When one refuses to write as part of the team, they mark themselves not just as an outcast from a social group, but as someone who is not included in the Scrum team.

This team based identity is shaped by a specific power structure that differs from how many people expect a workplace hierarchy to function. In a typical corporate discourse, there is often a very straightforward top-down power structure. A manager will assign a task and a subordinate will complete the task. Scrum replaces this with 3 roles: The Product Owner, the Scrum Master, and the Team. Within this framework, no single person has total control.

The Product Owner has the power of prioritization. They are allowed to decide what features should be included, and can order the product backlog (a list of all tasks

that need to be completed), but they do not get to decide how long it will take the team to build or how they should build it.

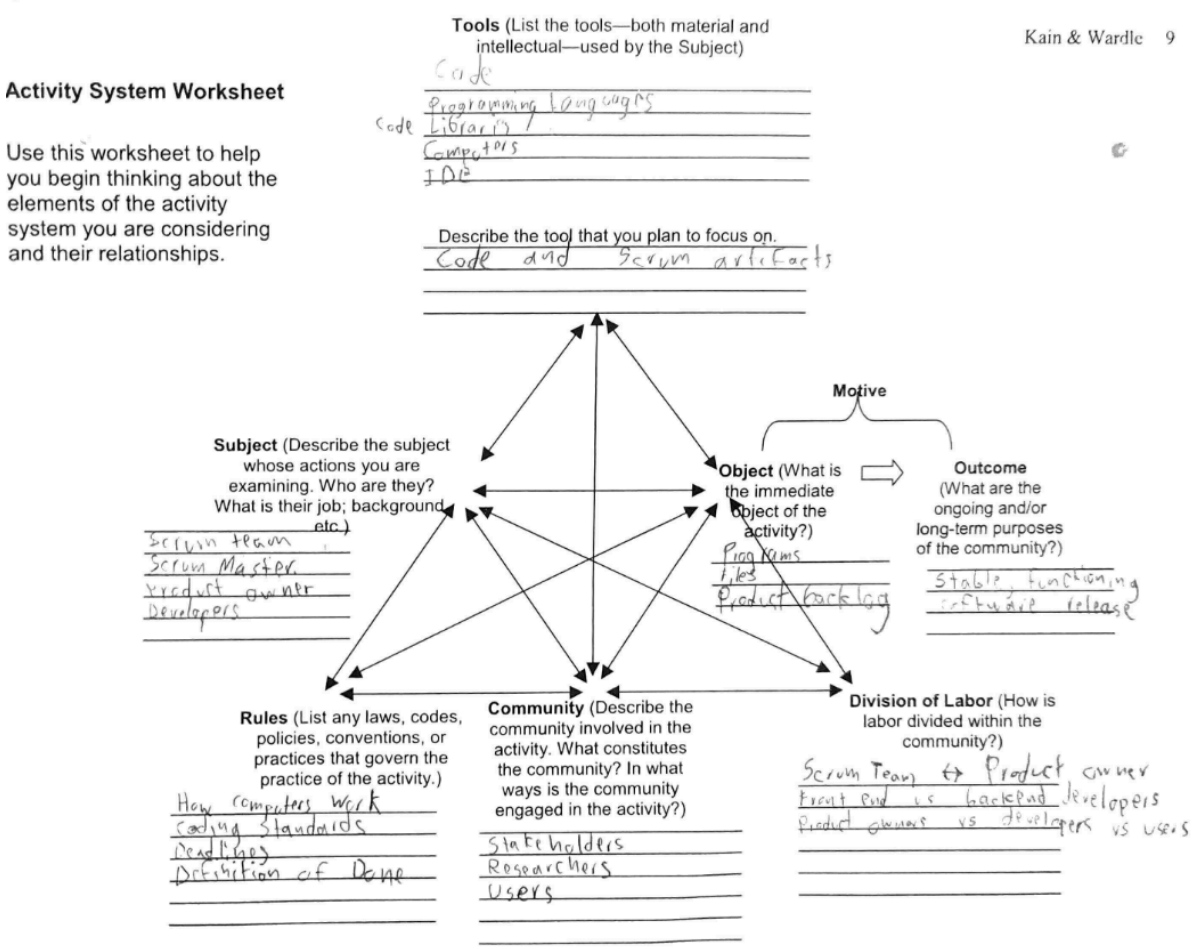
The development team is the party with the power to actually implement features. During a sprint planning meeting, the team pulls items from the product backlog (what needs to be done) to the sprint backlog (what is being done this sprint). This means the development team is the one that decides which work they will get done each sprint.

If the Product Owner controls the “what” and the development team controls the “how,” then the Scrum Master’s job is ensuring the team is adhering to all of the rituals of Scrum. They have no formal power to hire, fire, or assign tasks, and instead they are in charge of the “rules of Scrum.” Looking through Gee’s lens, the Scrum Master is the ultimate insider, and ensures all members are performing their identities “correctly.” Their primary task is often the removal of “blockers” (impediments to the developers, which are documented daily). The Scrum Master is the one who asks questions like “What is stopping us from meeting this sprint goal.” The Scrum Master’s power lies mostly in ensuring the Scrum framework stays intact and functioning.

Notably, within this distribution of power, each role relies on a specific type of writing to enact their authority. The Product Owner’s authority is written into the prioritized product backlog. By ordering items, they control the direction of the project without ever writing a line of code. Their writing takes the form of user stories and acceptance criteria, which define the ‘what’ in a concrete enough way for developers to act on, but abstract enough to leave the ‘how’ open to the developers. The development team, instead, enacts their authority through the technical writing, including code commits, pull requests, test cases, documentation, and more. Their writing is the

implementation that translates the Product Owner's vision into a real, functioning system. The Scrum Master's writing mostly takes the form of documentation: meeting notes, blocker logs, retrospective summaries, etc. Their authority is enacted over the Scrum process itself, rather than the product. In each case, writing is the mechanism through which authority is exercised.

While Gee's framework helps us understand the identities members take on, it is through the lens of Kain & Wardle's Activity Theory that we see how these identities translate to products being made. In a Scrum Team, writing and language are not just used to communicate information, but are also the primary tools for facilitating activity.



In this activity system, the Subjects (Product Owner, Developers, etc) use specific tools to achieve an objective. These tools include material objects like IDEs and compilers, but more importantly, they include discursive tools like programming languages and Scrum specific genres. The ultimate outcome is not just ‘code’, but a stable, functioning software release that will provide value to the end users. This activity system is also mediated by rules and the division of labor. The rules include everything from the rules of computers (their CPU architecture, speed, limitations) to socially constructed standards like coding standards, and the agreed upon ‘definition of done.’ Meanwhile, the division of labor ensures the product owner is focusing on the “what” while the developers control the “how.” When we view Scrum as an activity system like this, it becomes clear that writing is the primary tool that is used to move software development forward.

Like any activity system, Scrum is full of tensions and conflicts. As Kain and Wardle explain, activity systems are dialectically structured. This means aspects of the system are mutually dependent, and when one changes, others change in response, often in unanticipated ways (276). They also point out that conflicts within activity systems commonly arise around the division of labor and rules, as participants in the system may disagree about how work should be divided (276). Both of these dynamics can often be observed in a Scrum team. The most common source of conflict arises between the Product Owner’s ‘what’ and the development team’s ‘how.’ When a user story is ambiguous or when acceptance criteria are not clearly defined in writing, the team may build something that may not match the Product Owner’s vision, leading to wasted time and effort as well as frustration. Another common tension comes from the

‘definition of done’. When the rules about what counts as complete or not are not clearly written or applied, the developers and stakeholders can find themselves disagreeing about whether a sprint was successful or not. Lastly, there can be tension in the division of labor itself. A Scrum Master’s role as an enforcer of the process can conflict with developers who feel that excessive documentation and ritual is slowing them and therefore slowing the whole process down. In each of these cases, a breakdown is not happening in the code of the software, but in the writing that the activity system depends on to function. These conflicts illustrate the dialectical nature of the system. When a Product Owner changes a requirement mid-sprint (a change in the object) then the division of labor changes in response, the team’s timeline shifts, and the rules governing what counts as done may need to be renegotiated. Changes due to conflicts ripple through the entire system, which requires writings to be updated or further breakdowns will occur throughout the whole system.

By utilizing the frameworks of Swales’, Gee, and Activity Theory, it is clear that competence in this field is not just about knowing how to code. It is about navigating a complex discourse community where your ability to communicate is just as important as the code you create. The patterns I have observed in the Scrum team are not just found within software development. In any collaborative technical field, whether data science or open source development, the same dynamics can be seen. Specialized genres structure collaboration, written language signals membership and distributes authority, and breakdowns in communication lead to breakdowns in production. As a newcomer entering the field, I realize that becoming a computer scientist means mastering both the code, theory, and logic of computer science as well as the discourse communities

that give that work its meaning. Again, as my Instructor noted in our interview, without the writing to communicate a solution, the solution might as well not exist to anyone but you.

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